

Economic Analysis of Cargo Cycles for Urban Mail Delivery

Carine Choubassi, Dan P.K. Seedah, Nan Jiang, and C. Michael Walton

Freight constitutes a large portion of urban daily traffic, contributing to emissions, noise, and safety concerns. Moreover, urban freight logistics are often hindered by the last mile, characterized by significant delays and high delivery costs. Firms in dense urban areas are therefore seeking more efficient and reliable modes for their last-mile services. One mode that has been gaining widespread interest is the cargo cycle. Current research on cargo cycles is limited; however, their success in effectively delivering urban goods is gaining recognition. An examination was done of the economic feasibility of using different types of cargo cycles in varying urban contexts. A case study that assessed replacing the U.S. Postal Service vehicles with cargo cycles for last-mile mail deliveries in three population densities is presented. With the use of existing depots, the results of the case study indicate that electric cargo trikes have the lowest net present value among the modes in congested areas with high population densities such as central business districts. Moreover, having a depot positioned within the delivery area was found to play a significant role in increasing the competitiveness of trikes compared with other modes.

Urban freight delivery plays a crucial role in maintaining the vitality of an economy. On the basis of the Commodity Flow Survey of the U.S. Census Bureau and the Bureau of Transportation Statistics, 81% of goods in the United States originate or terminate in metropolitan areas, serving around 83% of the U.S. population (1). With the anticipated increase in population, the popularity of e-commerce, and the trends of millennials to move back to the cities, these values will most likely increase.

Despite the larger distances traveled by intercity freight, the last mile of freight delivery is the least efficient. According to the annual Urban Mobility Report by the Texas Transportation Institute (2012), “congestion in the nation’s largest urban areas cost the trucking industry \$33 billion in delay time and wasted fuel in 2009” (2). In 2008, 55% of U.S. empty truck miles were in metropolitan areas (1).

The lack of curbside space for legal parking and unloading is another common issue in urban areas. It results in truck cruising or in illegal truck parking, leading to lane blockages. Moreover, parking fines are a major cost incurred by delivery companies that have now begun including it as part of their regular budget for doing business (1). The result is an increase in the cost of operations.

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Freight delivery is also considered a major nuisance to urban citizens as it directly affects their quality of life, specifically in emissions, noise, and safety (3). Freight vehicles are often too large for the narrow urban streets they travel, thus affecting perceived safety, walkability, and aesthetics of neighborhoods.

The importance of efficient urban freight delivery has become obvious to policy makers and public officials. Several attempts are being made by governments around the world to address the environmental and social effects of urban deliveries. Some of the main objectives include reducing congestion, improving mobility, and controlling emissions and noise (3). Street design improvements that incorporate loading and unloading zones, enforced spatial and time restrictions on truck delivery, and intermodal logistics platforms are some solutions that have been implemented over the past decade (4).

One prominent proposition is the establishment of consolidation or distribution centers on the outskirts of major urban areas. However, the success of this technique relies on the use of smaller and possibly nonmotorized vehicles to carry out urban deliveries. In this context, the role of cargo cycles, commonly known as freight bikes or cargo bikes, is evident. Cargo cycles are in no way a new concept. Dating back to the nineteenth century, cargo cycles were primarily used by tradesmen, doctors, grocers, milkmen, messenger boys, public authorities (5), and service providers to manage business logistics in a cost-effective, efficient way (6). Their emergence as a solution in the urban freight sector, however, is only currently being recognized.

One of the main barriers hindering the shift to cargo-cycle deliveries is the lack of awareness among private industries about the costs incurred by this mode and its competitiveness in urban delivery. This paper attempts to model the cost-effectiveness of different types of cargo cycles in various urban contexts in comparison with other existing and emerging modes. For the remainder of this paper, the term cargo cycles will refer to any type of bicycle, tricycle, or quadricycle that is used to carry goods or passengers. Cargo bikes will exclusively refer to two-wheeled cargo cycles and cargo trikes will refer to three-wheeled cargo cycles.

The paper first presents an overview of cargo cycles and previous research that aims at assessing their effectiveness in the last-mile delivery of freight. The proposed model is then outlined, followed by a description of the case study through which it is validated. The results of the study are presented to demonstrate the competitiveness of cargo cycles and the policies and private-sector implications associated with their implementation.

BACKGROUND RESEARCH ON CARGO CYCLES

Trucks are the dominant mode of freight transport in many developed cities. Most of these trucks are often transporting lightweight goods and some are even empty. In the city of Breda in the Netherlands, 1,900 vans and trucks enter the city each day; 40% of the deliveries

involve just one box, and only 10% actually require a large vehicle to be transported (6). In other European cities, only 20% of all cargo trips require the use of heavy vehicles, and more than 12% are classified as small goods with an average volume of 1 m³ (7). The potential of replacing trucks with smaller modes such as cargo cycles is evident.

Potential of Cargo Cycles

Academic studies have been conducted to assess the competitiveness of cargo cycles in the delivery of urban freight. Based on research done by CycleLogistics, more than 25% of all goods and 50% of all light goods in European cities can be handled by cycles (8). A study done by Mühlbacher on the city of Graz, Greece, shows that 32% of kilometers driven could be transferred to cargo cycles which would lead to 1,200 tons of savings in fuel and 2,600 tons of savings in CO₂ (9). In Berlin, research indicates that cargo cycles with electric assist can replace up to 85% of car trips made by courier services (6).

The success of cargo cycles depends greatly on the urban context they are implemented in. Flat urban areas with high population (more than 100,000 inhabitants), high congestion levels, narrow streets, limited on-street parking, and a strong cycling community have proved to be the best conditions for the feasibility of this mode (6, 10). Tipagornwong and Figliozzi use an optimization-based cost-comparison model to show that the use of freight trikes in Portland makes the most sense in urban areas where parking or access to destinations is difficult, average speeds are relatively low, and there are time constraints for deliveries (11).

From an industry perspective, the cargo cycles' success is most pronounced in the delivery of light-to-medium-weight goods with an average payload of 100 to 1,000 lb and distances not exceeding 12 mi (12). Moreover, the presence of consolidation centers where goods are collected and redistributed within the urban area is found to be beneficial. Pilot studies in Europe and North America were conducted to assess the cost-effectiveness and efficiency of shifting from traditional delivery methods to electrically assisted modes. One example is the TNT Express demonstration in Brussels, which uses trucks as mobile depots. The truck would first deliver parcels to a central location in the city, and then deliveries are made with electric cycles (13).

Similarly, Conway et al. assess the feasibility of freight trikes in Manhattan, New York City, based on a thorough review of current operations in Paris and London (14). The results indicate that, with the availability of well-equipped microconsolidation centers, freight trikes are a viable option for freight delivery in Manhattan. Similarly, a study done by Outspoken Delivery, the largest cycle courier in the United Kingdom, in partnership with the CYCLElogistics project, highlights the importance of consolidation hubs on the edge of urban areas (10). A study by Browne et al. further validates this point with results showing that urban microconsolidation centers coupled with the use of electrically assisted cargo tricycles leads to a 20% decrease in total distance traveled and a 54% decrease in CO₂ equivalent emissions in the city of London (15). The study also shows that, although shifting to cargo cycles leads to a slight increase in labor cost, the total operating cost incurred would remain the same as for diesel-based delivery systems.

Advantages of Cargo Cycles

On the basis of several experiences in the adoption of cargo cycles in urban freight logistics, a number of advantages can be outlined.

In regard to cost-effectiveness, cargo cycles can be significantly less expensive per unit of distance than other alternatives (16). This is a result of their low capital, operating, and maintenance costs owing to fuel savings, lower insurance costs, and minimal need for storage. In addition, cargo cycles are prominent for their ability to enter pedestrianized neighborhoods and narrow alleyways, so they are highly reliable and resistant to uncertain road conditions such as congestion and unavailability of parking.

From a socioeconomic perspective, cargo cycles contribute to major decreases in unemployment, since they do not oblige employees to own a driver's license, and that increases the industry hiring pool (6). A study performed in India by Sadhu et al. assesses the benefits associated with cycle rickshaw trolleys (CRT), the most widely used nonmotorized mode for intracity freight transport in Delhi, through 2000 surveys with Delhi-based CRT drivers (17). The study concludes that one of the most significant advantages of this mode is the number of jobs it creates, which can be up to 700,000 jobs.

From an environmental perspective, cargo cycles are considered cleaner and quieter as they do not emit any local air or noise emissions (18). Sadhu et al. confirm that CRTs are associated with 3% of CO₂ savings and an 8% reduction in hydrocarbon emissions. Moreover, shifting to CRT leads to a major decrease in the number of vehicle-related accidents and fatalities, thus improving roadway safety (17).

Disadvantages of Cargo Cycles

In contrast, cargo cycles also have some limitations. Electric batteries are hindered by low travel ranges per charge (25 to 40 mi), thus compelling trips to be shorter (7). A cyclist's age and physical fitness may impose restrictions and entail higher labor costs, as well as higher health insurance and benefits costs owing to the more arduous nature of the job. Weather conditions and topography can also affect the feasibility of this mode, which may not be well adopted in icy environments or in hilly regions.

For governments and local authorities, the introduction of cargo cycles requires adequate planning and budget allocations to ensure consistent maintenance and management of cycling infrastructure, modification or creation of certain legal frameworks associated with this new mode, and enforcement of proper cargo cycle-related regulations that target pedestrian safety issues (8).

With rising awareness, more innovative designs, and successful tests being carried out, the use of cargo cycles in urban freight transport is expected to snowball. Detailed analyses of the cost-effectiveness of cargo cycles in different urban contexts are therefore needed. Despite the growing interest in this mode, there seems to be a major gap in research and data collection on its actual competitiveness and performance. Most studies rely on surveys and personal interviews, and only a few try to incorporate quantitative assessments.

This paper presents a cost comparison of cargo cycles and other modes in one of the fastest-growing U.S. cities, Austin, Texas. That city is characterized by a medium-dense population, well-maintained and expanding bicycle infrastructure, and strong cycling community. The industry under study is the U.S. Postal Service (USPS), the nation's federal postal service provider, which has so far relied on its customized Long Life Vehicles (LLV). The paper examines the economic benefit of shifting from these vehicles to cargo cycles in three urban contexts within the city and in varying traffic conditions.

METHODOLOGY

Data and Assumptions

The USPS case study is interesting for several reasons. First, USPS Every Door Direct Mail (EDDM) service (19) constitutes a good data source to estimate the number of deliveries that can be made in a given zip code. Second, USPS has recently declared its desire to replace its currently owned fleet of vehicles and has called for bids for a new type of vehicle. Any changes in vehicle type will have long-term implications on its operations and consumer footprint.

Purchased between 1987 and 1994, the Grumman LLV has been USPS's main vehicle for the past few decades, constituting 74% of its entire vehicle fleet (20). Currently, 143,000 LLVs are operating in the United States, and many are in poor condition, requiring high maintenance costs. USPS has undertaken a number of attempts since 2005 to replace, repair, or modify its aging fleet (21).

For the purpose of comprehensively assessing cargo cycles, two types are considered: the cargo bike and the cargo trike. Moreover, two types of electric propulsion systems are analyzed: pedelecs, which are equipped with small motors that produce power only when the rider is pedaling, and electric cycles (e-cycles), which are equipped with batteries that can operate during the entire trip duration with no need for pedaling. It is assumed that the pedelecs have a maximum

operating speed of 10 mph, while the e-cycles can achieve a speed of 15 mph. Regular nonmotor-assisted cargo cycles are disregarded in this study to downplay limitations associated with topographical issues and driver fatigue.

The cost of the cargo cycles is compared with that of the LLV and the electric van, a new vehicle that relies solely on electric propulsion and has been recently introduced to the courier market (22). To achieve a realistic representation, the exact make and model for each mode are assumed, as well as the corresponding purchase costs and operating and maintenance costs for each. Table 1 summarizes these characteristics.

On the basis of USPS requirements, delivery vehicles often travel at an average speed of 15 mph owing to frequent stops (23). It is assumed that all modes have to travel at an average speed equal to or less than 15 mph to be able to provide adequate delivery services.

In regard to location characteristics, the case study is performed on three zip codes within the city of Austin. The model is applied to each zip code separately, and then the results are compared. The locations considered are as follows:

- Austin central business district (CBD), zip code 78701: characterized by offices and commercial buildings with an area of 1.6 mi² (3,632 people/mi²) and 85 deliveries/mi.

TABLE 1 Characteristics of Alternatives (23–35)

Characteristic	LLV	Van	Pedelec Bike	E-Bike	Pedelec Trike	E-Trike
Manufacturer	Grumman Corporation	Nissan	Larry vs. Harry	Larry vs. Harry	Cycles Maximus	Cycles Maximus
Make	LLV	Nissan e-NV200	Bullit Cargobike	Bullit Cargobike with Bionx Conversion Kit	Pedal Cargo Van Trike	Traction Drive Cargo Van Trike
Life span (years)	24 ^a	20 ^b	5 ^c	5	5	5
Single charge range (mi)	na	106 ^b	75 ^d	25 ^c	90 ^d	30 ^e
Maximum operating speed (mph)	55 ^a	75 ^b	10 ^f	15.5 ^g	10	15.5
Volume capacity (ft ³)	121 ^a	148 ^b	5	5 ^c	49	49 ^e
Payload (lb)	1,000 ^a	1,697 ^b	200 ^h	200 ^h	550 ^e	550 ^e
CO ₂ emissions (g/mi)	1,267	194 ⁱ	35 ^j	35	35	35
Purchase cost (\$/vehicle)	21,538 ^k	32,301 ^b	1,244 ^c	4,058 ^c	6,491 ^e	11,236 ^e
Fuel or electric motor charge cost (\$/mi)	0.13 ^l	0.05 ^b	0.05	0.05	0.05	0.05
Maintenance cost (\$/mi)	0.15 ^m	0.1174 ⁿ	0.02 ^o	0.02	0.02	0.02
Labor cost (\$/h)	28 ^p	28	28	28	28	28
Emissions cost (\$/mi)	0.0168 ^q	0.00257 ^r	0.000468 ^j	0.000468	0.000468	0.000468

NOTE: na = not applicable.

^aUSPS LLV specs (23).

^bNissan e-NV200 specs (24).

^cLarry vs. Harry Bullit cargobike specs (25).

^dAssumed equal to three times that of an electric battery.

^eCycles Maximus CM Traction Drive specs (26).

^fJensen et al. (27).

^gRolling orange bikes (28).

^hPayload is 400 lb, including weight of driver. Driver assumed to be another 200 lb.

ⁱAssumed equal to that of a Nissan Leaf.

^jEuropean Cyclists Federation (29).

^kU.S. Government Accountability Office (30).

^lAssumed equal to that of a pickup/van/SUV in city driving conditions according to *The Per-Mile Costs of Operating Automobiles and Trucks* values (31).

^mSeedah et al. (32).

ⁿAssuming each van is driven 40 mi/day on average for 7 years.

^oTipagornwong and Figliozzi (11).

^pEach vehicle is assumed to be driven by one employee paid by the hour. The business hours of each location depend on the service hours of its respective depot.

^q11.52 kg of CO₂/gal of gasoline and 0.11 gal/mi with \$12 CO₂/ton (33).

^rAssumed equal to that of a Nissan Leaf, as estimated by the U.S. Environmental Protection Agency report (34).

- University of Texas at Austin (UT) neighborhood, zip code 78705: characterized by a mixture of commercial and residential neighborhoods surrounding UT. The 2.2-mi² area has a population density of 14,246 people/mi² and around 82 deliveries/mi.

- Far West neighborhood, zip code 78731: mainly residential and encompasses many single-family homes, some multifamily homes, and stretches of open space. The zip code covers an area of 8.6 mi², has a population density of 2,977 people/mi², and 35 deliveries/mi.

For the remainder of this paper, the three locations will be referred to as CBD, UT neighborhood, and Far West, respectively. The choice of locations provides two classifications of urban settings: high-density (CBD and UT neighborhood) and low-density (Far West), as well as three types of depot or consolidation center locations, as shown in Figure 1. Depots are assumed to be located where the USPS offices that service each zip code are located.

For the CBD location, it is assumed that there are two depots, a main depot external to the zip code and a local depot located within the given zip code. The assumption is that for the LLV and van, this location is serviced only by the main depot, in which the vehicles are required to travel a distance of 3.7 mi before they reach the area where deliveries have to be made. This assumption is made because no parking lots were found to accommodate the LLV and van in the vicinity of the local depot. However, for the cargo cycle alternatives, it is assumed that the vehicles are parked at the local depot. The goods are first transported from the main depot via a 15-ft truck (incurring an additional cost of \$10,099) and then distributed to the zip code locations following cargo cycles. For these alternatives, it is assumed that the truck has to first travel a distance of 4.7 mi from the main depot to the local depot and unload, and then all internal deliveries are made using cargo cycles.

For the UT Neighborhood, it is assumed that there is only one depot, a main depot located 0.5 mi away from the zip code. As for the Far West location, it is also equipped with one depot located within the zip code where all delivery trips originate.

Analysis Method

Scenario Assessment

The model assumes that the entire fleet of LLVs is replaced with a uniform fleet of a different type of vehicle, with the base case being a new fleet of LLVs. Each mode is therefore assessed separately. The different alternatives are compared in light of two traffic conditions (congested and normal). It is assumed that the average traffic speeds on the roadway are 10 mph during congestion and 15 mph during normal conditions. The cycle modes are considered to have their own dedicated lanes and are not affected by congestion on the roadway. The following assumptions result in six scenarios assessed in congested and normal conditions:

1. Base case: Purchase new fleet of LLVs (10 mph and 15 mph);
2. Replace LLV with van (10 mph and 15 mph);
3. Replace LLV with pedelec bike (10 mph);
4. Replace LLV with e-bike (15 mph);
5. Replace LLV with pedelec trike (10 mph); and
6. Replace LLV with e-trike (15 mph).

The aforementioned scenarios are then compared in each of the three locations separately, which results in 36 scenarios.

Number of Deliveries

By using the city of Austin's geographic information system data (35) and the EDDM routes defined by USPS, each link is assigned a total number of deliveries based on a proportional percentage of the number of deliveries on its corresponding carrier route, as shown in Equation 1.

$$q_{ij} = \left[\frac{l_{ij}}{\sum_{k=1}^n l_{ik}} \right] Q(R_i) \quad (1)$$

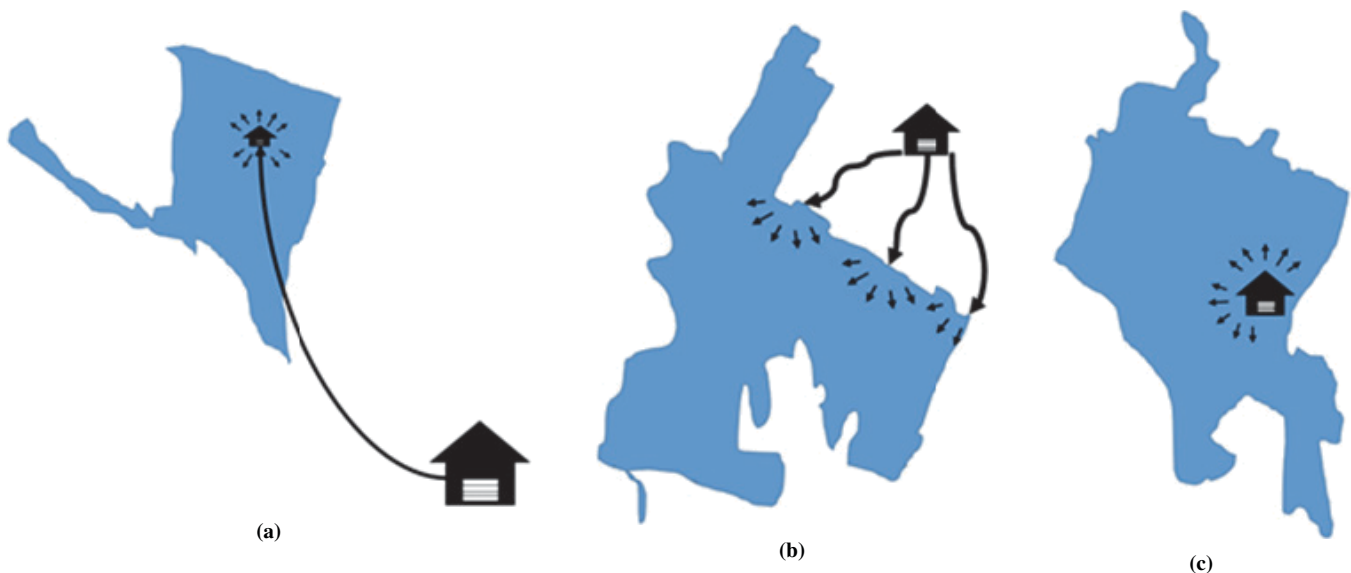


FIGURE 1 Depot configuration for each of the locations: (a) CBD, (b) UT neighborhood, and (c) Far West.

where

q_{ij} = total number of deliveries on link (i, j) ;

l_{ij} = length of link (i, j) (mi);

n = total number of links in route R_i ; and

$Q(R_i)$ = total number of deliveries on carrier route R_i .

Vehicle Capacity

On the basis of the EDDM package specifications, the maximum allowable dimensions are 15 in. $\times$ 12 in. $\times$ 0.75 in., resulting in a package volume of 135 in.³ (0.078125 ft³). Using this estimate, the number of deliveries (total number of packages of this size) that can be made by each mode during one trip, also defined as the capacity of this alternative (Q_{Alt}) expressed as deliveries per trip, can therefore be calculated based on Equation 2.

$$Q_{Alt} = \left(\frac{\text{number of deliveries}}{\text{trip}} \right) = \frac{Q_{Alt} \left(\frac{\text{ft}^3}{\text{trip}} \right)}{0.078125 \frac{\text{ft}^3}{\text{delivery}}} \quad (2)$$

This results in a capacity of 1,549 deliveries per trip for the LLV, 1,894 deliveries per trip for the van, 64 deliveries per trip for the bike, and 627 deliveries per trip for the trike.

Network Model

The network problem can be classified as a capacitated vehicle routing problem with time window (i.e., CVRPTW). All deliveries within a given location are required to be made during a 1-day working schedule (~ 8 h). Moreover, each type of vehicle has its own set of optimum carrier routes with minimum trips lengths and travel times. Based on these criteria, the required number of vehicles for each alternative is determined optimally. The model takes capacity, route length, and time window as constraints. In this study, the time window is based on the business hours of each location depot (36). The problem can then be stated as follows:

$$\text{minimize } \sum_{k \in K} \sum_{(i,j) \in A} c_{ij} x_{ij}^k \quad (3)$$

subject to

$$\sum_{k \in K} \sum_{j \in \delta^+(i)} x_{ij}^k = 1 \quad i \in N \quad (4)$$

$$\sum_{j \in \delta^+(0)} x_{0j}^k = 1 \quad k \in K \quad (5)$$

$$\sum_{j \in \delta^-(j)} x_{ij}^k - \sum_{j \in \delta^+(j)} x_{ji}^k = 0 \quad k \in K; j \in N \quad (6)$$

$$\sum_{j \in \delta^-(n+1)} x_{jn+1}^k = 1 \quad k \in K \quad (7)$$

$$x_{ij}^k (w_i^k + s_i + t_{ij} - w_j^k) \leq 0 \quad k \in K; (i, j) \in A \quad (8)$$

$$\alpha_i \leq w_i^k \leq b_i \quad k \in K; i \in V \quad (9)$$

$$\sum_{i \in N} q_i \sum_{j \in \delta^+(i)} x_{ij}^k \leq Q \quad k \in K \quad (10)$$

$$x_{ij}^k \in \{0, 1\} \quad k \in K; (i, j) \in A \quad (11)$$

where each of the variables is defined as follows:

A = arc set representing all paths in the network;

N = set of nodes ($N = V$);

K = set of m vehicles of fixed fleet available at the depot;

c_{ij} = travel cost of each route (i, j) ;

x_{ij}^k = binary variable with $(i, j) \in A$; $k \in K$. It is equal to 1 if and only if arc (i, j) is used by vehicle k ;

w_i^k = continuous variable with $i \in N$; $k \in K$, indicating the time at which vehicle k starts servicing vertex i ;

s_i = service time at i (with $s_0 = s_{n+1} = 0$), based on a 5-s per package assumption;

t_{ij} = travel time from i to j ;

a_i = lower limit of time window for vertex i ;

b_i = upper limit of time window for vertex i ;

V = vertex set representing customers with $V = 0$ and $V = n + 1$ representing depot;

q_i = demand of customer i ;

Q = capacity of each vehicle;

$\delta^+(i)$ = with $\delta^+(i) = \{j: (i, j) \in A\}$; and

$\delta^-(j)$ = with $\delta^-(j) = \{i: (i, j) \in A\}$.

The objective function (Equation 3) minimizes the total routing cost. The constraint in Equation 4 states that each customer is visited exactly once, while constraints in Equations 5 through 7 ensure that the flow conservation is satisfied at each customer vertex. The consistency of the time variables w_i^k is ensured through the constraint in Equation 8, while time windows are imposed by the constraint in Equation 9. These constraints also eliminate subtours. Finally, the constraint in Equation 10 enforces the vehicle capacity restriction.

For each zip code, the model determines the number of trips needed to perform all deliveries within a given time window such that every customer is visited only once, deliveries per trip do not exceed capacity of the vehicle, all vehicles originate and terminate at the depot, and total trip distances do not exceed the maximum allowable range of electric motors.

The solution algorithm used in this study is an all-purpose heuristic based on the ruin and recreate principle, created by Schrimpf et al., which is suitable for complex optimization problems with many constraints (37). An initial CVRPTW solution is created, then a part of it is significantly ruined (or eliminated), and then it is recreated in later stages of the process. By performing several iterations of this procedure, a better solution is obtained (38). The main advantage of this methodology is that, instead of creating a new solution from scratch, it tries to arrive at the best solution by fixing an existing one (37).

With the use of jspritMATSim (39), the software application generates a plot diagram for each scenario being assessed. The plot of the LLV in normal conditions for the UT Neighborhood location is presented in Figure 2.

Total trip time is determined as the sum of the trip duration and a certain loading time, which is also assumed to be 5 s per package. The number of vehicles is then determined by using the best combination algorithm (40), which finds the best combination of trips that can be allocated to a single vehicle such that the time window constraint or

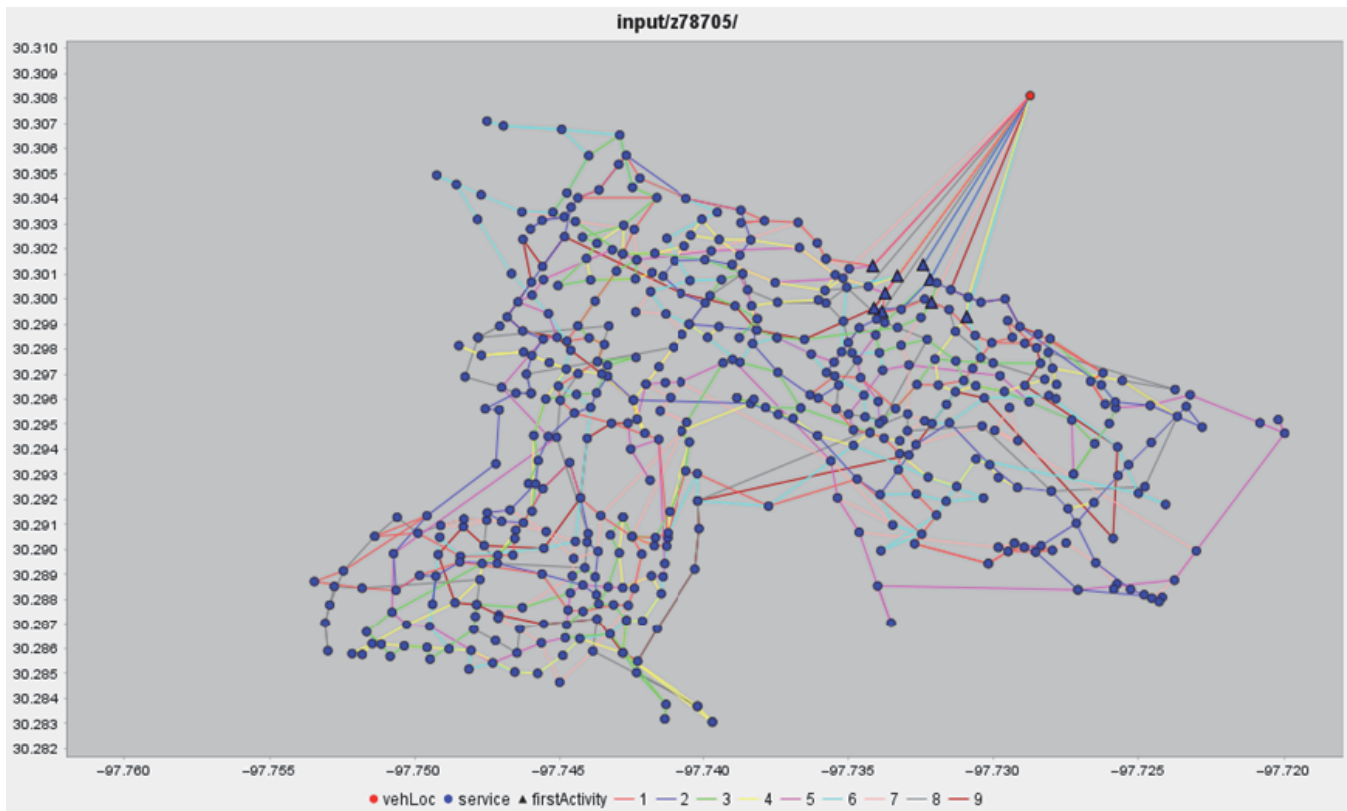


FIGURE 2 Plot diagram of LLV in UT Neighborhood normal conditions.

maximum allowable range, or both, are satisfied. Consequently, the optimum number of vehicles needed to make all deliveries in a zip code during 1 business day can be determined and inputted in the following financial model.

Financial Model

The four vehicle types are compared by using the net present value (NPV) of the estimated total long-term cost associated with each. This is done by developing a financial cost model for a period of 24 years (equal to the lifetime of the LLV) starting with year 2015 as the base year. A discount rate of 6% is used as the analysis is considered to be of the relatively lower-risk magnitude (41). The model is constructed by using the monetary value of the purchase cost, salvage value, fuel or electric motor charge cost, maintenance cost, labor cost, emissions cost, and battery-replacement cost for each of the vehicles.

DISCUSSION OF RESULTS

Analysis of Results

With the use of the generated NPV values, the costs of the modes can be compared. Since the resulting NPV values portray cost incurred, then the more negative NPV value (higher number in absolute value) indicates a higher cost for the analysis period. The mode with the least negative NPV value (lowest absolute value) is considered the least expensive and therefore the best option according to this analysis.

The results shown in Figure 3 indicate that the pedelec bike and the e-bike have the highest NPV for all scenarios. Moreover, their NPV values are 40% to 120% higher than those of the trikes regardless of traffic conditions and location. As a result, this type of cargo cycle is rendered unfeasible from a cost perspective and will from now on be disregarded in this analysis.

The true competitiveness of the cargo cycles is portrayed through the NPV values of the trikes. In the following analysis, two main parameters will be analyzed: the NPV of the trike, which portrays its cost; and the percentage difference among the NPV of the trikes and those of the van and LLV. These values will be compared among locations, as well as traffic conditions to determine the potential effect of each on the cost and competitiveness of the trike.

If one looks at the total NPV cost values of each mode separately, it appears that the e-trike yields the lowest NPV cost in congested conditions in the CBD location among all the scenarios. Moreover, the e-trike has the lowest NPV in congested conditions and is therefore the most cost-effective among all modes regardless of location. This leads to the conclusion that for the CBD location, during congested conditions, the e-trike is less expensive to use than the LLV and the van. As for other locations and traffic conditions, the trike incurs a higher NPV cost, yet the percentage difference in its cost compared with those of the LLV and van is generally small. As a result, it is important to also assess the competitiveness of this cargo cycle in different contexts.

If one defines the competitiveness of the trike as the percentage difference between its own NPV and that of other modes, then the lesser this difference, the more competitive is the trike. On the basis of the results obtained, several conclusions can be made. The

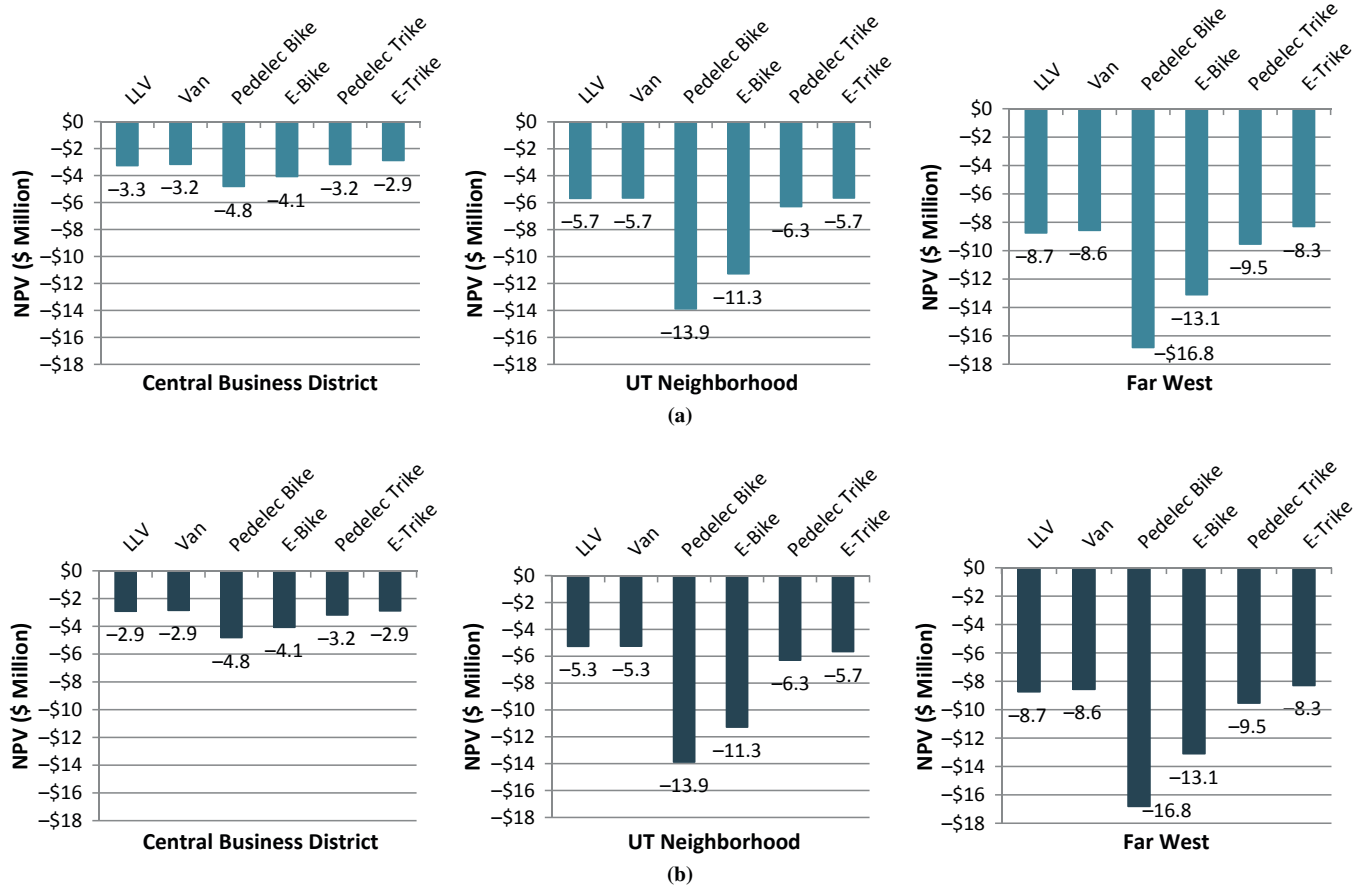


FIGURE 3 NPV (\$ million) of total cost for all scenarios: (a) contested conditions and (b) normal conditions.

trikes turn out to be more competitive in congested traffic conditions than in normal conditions regardless of location. This is under the assumption that the trikes are not affected by traffic conditions and have access to dedicated bike lanes. Comparing the three zip codes shows that the trikes are, in general, more competitive in the CBD than in the UT Neighborhood, which suggests that having the depot within the delivery area improves the efficiency of this mode. This can be justified by the additional number of trips the trikes have to make (owing to lower capacity) compared with other modes. Consequently, allocating the depot farther away from the delivery areas adds an additional distance for the trikes to travel during each trip to and from the depot. Longer distances increase the cost of the trikes and make them less competitive when compared with other modes that can accommodate longer trips.

The comparison between the Far West neighborhood and the CBD suggests that the trikes are more competitive in areas with higher deliveries per mile (higher population density). This can be justified by considering the number of deliveries per mile per vehicle, which shows that as that number increases, the NPV of the trike decreases (in absolute value). In regions that are more spread out such as the Far West neighborhood, the trikes are required to travel longer distances to make the same number of deliveries. This results in a higher total distance traveled (total miles) and therefore a higher cost.

Assuming that there is no dedicated right-of-way for the e-trikes and that they are required to share the road with the other vehicles, then only the assumptions in congested conditions would change. Under this scenario, the e-trikes will have to travel at a speed of

10 mph (as opposed to 15 mph in their own dedicated lane) at a speed equal to that of the LLV and van. This results in a scenario comparable with the pedelec scenario in congested conditions (where all vehicles are traveling at a speed of 10 mph), and this is expected to entail an even higher cost for the e-trike and makes it less competitive. This observation suggests that the provision of a dedicated right-of-way for the e-trikes would decrease their cost and increase their competitiveness.

Policy Implications

Results of the proposed model outline a number of factors that increase the competitiveness of cargo cycles in urban areas. From a policy standpoint, such factors should be promoted and made available to encourage courier and logistics companies to shift to cycle deliveries. The economic feasibility of cargo cycles is more pronounced with the presence of bike lanes and local depots close to or within the delivery areas. Such facilities are equipped with loading and unloading zones and can be provided by local governments in designated areas and leased to one or more logistics companies.

The comparison between different types of urban areas also shows that the use of cargo cycles is most successful in regions with high population density such as CBDs. Such areas also play a major role in the economy of cities, for they account for a significant percentage of employment within metropolitan areas, and they are often composed of mixed-use neighborhoods. Policy makers should therefore start by focusing on the CBDs of large, dense cities, where

current delivery methods are hindered by congestion and lack of parking space.

The assignment of no-truck zones within certain CBD streets is another contributing factor. Facilitating access of trucks to central hubs and denying them access to narrow residential streets could play a major role in this shift. Similar policies include higher parking fines or monetary incentives to shift to cargo cycles.

Most important, governments should support and advocate the use of cargo cycles for reasons not assessed in this study, such as the improvement of safety and environmental conditions. Other benefits include improving the walkability score of a CBD and the overall livability of the area, thus improving the quality of life of urban citizens.

Private-Sector Implications

From a private-sector perspective, the model outlines the feasibility of the use of cargo cycles in terms of total cost. With the availability of adequate bike lanes, as well as proper consolidation centers, it is expected that logistics companies would decrease their cost of service by switching to this alternative mode; that could also allow them to lower their price of service and become more competitive in the market.

Moreover, the use of cargo cycles portrays a more environmentally friendly image for private companies, making them more appealing to customers. For logistics companies that offer delivery services exclusively, the cargo trikes can provide a platform for advertisement that has proved to generate a good source of revenue that can be used to increase hourly wage rates of employees.

Recommendations and Future Work

The proposed methodology is applied to one city and can be further applied to other U.S. cities with different characteristics with corresponding USPS EDDM data. The financial analysis of the model can also be enhanced to include other costs not accounted for in this study, such as insurance, registration, and depot operation. Moreover, modifying the model to include some of the nonquantifiable benefits associated with cargo cycles would further portray their competitiveness. The model also assumes that local depots and bike lanes are available. This, however, may not be the case in most U.S. cities today, and it may prove arduous and costly to implement. It would be useful to incorporate the cost of constructing such facilities and the ways to optimize these costs.

In regard to network modeling, the current methodology relies on distances, speeds, and number of deliveries. Other constraints could be incorporated, such as no-truck zones; streets with prohibited parking; and the effect of topography, weather, and driver fatigue. Another suggestion is to evaluate the use of mixed fleets to determine the cost of an optimum fleet combination of all the proposed modes. Other policies could also be considered, such as overtime hours as opposed to business hour restrictions.

CONCLUDING REMARKS

With growing interest in cargo cycles, some academic studies assess their feasibility in urban settings. This paper focuses on the economic feasibility and competitiveness of cargo cycles with respect

to other existing and emerging modes in different urban contexts. A case study on the USPS EDDM service in the city of Austin is presented.

The results seem to confirm findings of previous studies. From a cost perspective, the most cost-effective mode for urban deliveries in the CBD in congested conditions is the e-trike. Moreover, the e-trike has proved to be relatively competitive with other modes even in cases where its costs are higher. Competitiveness of the trikes varies based on the urban context and is highest in areas with high population density and with depots located within the delivery area boundaries. The results also show that, as the deliveries per mile required by a single trike increase, the cost of the trikes decreases greatly.

The outlined methodology demonstrates the feasibility of cargo cycles from the cost perspective of the private companies. However, it assumes that proper facilities such as dedicated bike lanes and consolidation centers are already available within the urban areas. This suggests that successful implementation of this mode is also dependent on the efforts of the public sector in providing the required amenities. Moreover, policies that discourage the use of trucks and motorized vehicles play a major role in further promoting the shift to cargo cycles.

The benefits of cargo cycles for the quality of life of urban citizens are evident. Increased livability, improved safety, decreased congestion, and reductions in noise and air emissions are some examples. This study shows that the shift to cargo cycle urban deliveries would result in a win-win scenario for both public and private sectors. With proper awareness and cooperation, this mode has a high potential in leading future urban logistics and creating cleaner and more livable environments in the nation's growing cities.

REFERENCES

1. Rhodes, S. S., M. Berndt, P. Bingham, J. Bryan, T. J. Cherrett, P. Plumeau, and R. Weisbrod. *NCFRP Report 14: Guidebook for Understanding Urban Goods Movement*. 2012. http://onlinepubs.trb.org/onlinepubs/ncfrp/ncfrp_rpt_014.pdf.
2. TTI. 2012 Urban Mobility Report. 2012. <http://d2dti5nnlpr0r.cloudfront.net/tti.tamu.edu/documents/mobility-report-2012.pdf>.
3. Crainic, T. G., N. Ricciardi, and G. Storchi. Advanced Freight Transportation Systems for Congested Urban Areas. *Transportation Research Part C*, 2004, pp. 119–137. http://ac.els-cdn.com/S0968090X04000142/1-s2.0-S0968090X04000142-main.pdf?_tid=28d4a288-ea07-11e4-b139-00000aacb362&acdnat=1429827899_474a87049fdae010013950b7ba169016.
4. Munuzuri, J., J. Larraneta, L. Onieva, and P. Cortes. Solutions Applicable by Local Administrations for Urban Logistics Improvement. *Cities*, Vol. 22, No. 1, 2005, pp. 15–28. http://ac.els-cdn.com/S0264275104001313/1-s2.0-S0264275104001313-main.pdf?_tid=d2eef8cc-ea07-11e4-a1f8-00000aacb0f26&acdnat=1429828185_fd517d19c02415aa313b9712c494def2.
5. Klepfer, M. *The Cargo Bike: A Vehicle That Will Change Your Life*. July 2012. <http://momentummag.com/the-cargo-bike-a-vehicle-that-will-change-your-life/>.
6. Decker, K. D. *Cargo Cyclists Replace Truck Drivers on European City Streets*. Sept. 2012. <http://www.lowtechmagazine.com/2012/09/jobs-of-the-future-cargo-cyclist.html>.
7. Jorna, R., and M. Mallens. *Promoting Electric Bikes and Scooters for Delivery of Goods and Passenger Transport in Urban Areas*. Mobycon, Delft, Netherlands, 2013.
8. CycleLogistics—Moving Europe Forward. 2011. http://cyclelogistics.eu/docs/111/D2_1_Analysis_of_Cargo_Cycling_v_2_Sept2013.pdf.
9. Mühlbacher, S. *Bicycle Logistic: Potential for Bicycle Deliveries in Graz*. Diploma thesis. FH Joanneum, Graz, Switzerland, 2010. https://www.fh-joanneum.at/aw/home/Studienangebot_Uebersicht/departement_bauen_energie_gesellschaft/evu/Menschen/AbsolventInnen/~bifl/Abschlussarbeiten_Details_evu/?arbid=4304607389&arbtyp=Dipl%2E-Arbeit&lan=en.

10. Outspoken Delivery. *Loads More Cycling*. 2012. <http://www.ctc.org.uk/file/public/cargo-cycling.pdf>.
11. Tipagornwong, C., and M. Figliozzi. Analysis of Competitiveness of Freight Tricycle Delivery Services in Urban Areas. In *Transportation Research Record: Journal of the Transportation Research Board*, No. 2410, Transportation Research Board of the National Academies, Washington, D.C., 2014, pp. 76–84.
12. Riehle, E.-B. *Cargo Bikes as Transportation Vehicles for Urban Freight Traffic*. MS thesis. TU Dortmund University, DLR Institute of Transport Research, Dortmund, Germany, 2012. http://www.ecf.com/wp-content/uploads/master-thesis-Ernst-Riehle_synopsis.pdf.
13. TNT Express. Demonstration B: TNT Express in Brussels—City Logistics Mobile Depot. STRAIGHTSOL Project Demonstrations, Feb. 28, 2012. http://straightsol.eu/demonstration_B.htm. Accessed Nov. 5, 2015.
14. Conway, A., P.-E. Fatisson, P. Eickemeyer, J. Cheng, and D. Peters. Urban Micro-Consolidation and Last-Mile Goods Delivery by Freight-Tricycle in Manhattan: Opportunities and Challenges. Presented at 91st Annual Meeting of the Transportation Research Board, Washington, D.C., 2012.
15. Browne, M., J. Allen, and J. Leonardi. Evaluating the Use of an Urban Consolidation Centre and Electric Vehicles in Central London. *IATSS Research*, Vol. 35, No. 1, July 2011, pp. 1–6.
16. Dembosky, A. Unburdened by Gas Costs, Bike Couriers See a Chance. *New York Times*, Sept. 2008, p. 5. [http://www.lexisnexis.com/hottopics/inacademic/?verb=sr&csi=6742&sr=HLEAD\(Unburdened%20by%20Gas%20Costs,%20Bike%20Couriers%20See%20a%20Chance.\)%20and%20date%20is%202008](http://www.lexisnexis.com/hottopics/inacademic/?verb=sr&csi=6742&sr=HLEAD(Unburdened%20by%20Gas%20Costs,%20Bike%20Couriers%20See%20a%20Chance.)%20and%20date%20is%202008).
17. Sadhu, S. L. N. S., G. Tiwari, and H. Jain. Impact of Cycle Rickshaw Trolley (CRT) as Non-Motorised Freight Transport in Delhi. *Transport Policy*, Vol. 35, May 2014, pp. 64–70. http://ac.els-cdn.com/S0967070X14001140/1-s2.0-S0967070X14001140-main.pdf?_tid=7acda95c-ea18-11e4-8f9c-00000aab0f02&acdnat=1429835338_030a90234e6d4ad623959928a75f8ab4.
18. ECLF. Guidelines Factsheet Cargo Bikes. Berlin, 2009. http://cyclelogistics.eu/docs/Guideline_-_Municipal_use_of_cargo_bikes_Friends_of_the_Earth_Germany.pdf.
19. *Advertise with Every Door Direct Mail*. U.S. Postal Service, 2015. <https://www.usps.com/business/every-door-direct-mail.htm>.
20. Sheehan, B. Pushing the Envelope: The USPS Long Life Vehicle. *Postal News*, Jan. 22, 2015. <http://postalnews.com/blog/2015/01/22/pushing-the-envelope-the-usps-long-life-vehicle/>.
21. O'Keefe, E. Postal Service in a Bind on Upkeep of Vehicle Fleet. *Washington Post*, June 18, 2010.
22. Edelstein, S. U.S.P.S. "Long Life" Vehicles Last 25 Years, But Age Shows Now. *Green Car Reports*, Feb. 17, 2015. http://www.greencarreports.com/news/1096830_u-s-p-s-long-life-vehicles-last-25-years-but-age-shows-now.
23. *Delivery Vehicle Fleet Replacement*. Management Advisory Report. U.S. Postal Service, June 10, 2014. <https://www.uspsoig.gov/sites/default/files/document-library-files/2014/dr-ma-14-005.pdf>.
24. e-NV 200. Specs. Nissan, 2015. http://www.nissan.co.uk/dam/services/gb/brochure/Nissan_e-NV200_van_Technical_Specs_UK.pdf.
25. Larry vs. Harry. Bullitt. Larry vs. Harry, 2013. <http://shop.larryvsharry.com/full-bike.html>.
26. Cycles Maximus. Electric Assistance for our Trikes. Cycles Maximus Sustainable Transport Solutions, 2015. <http://www.cyclesmaximus.com/electricassist.htm>.
27. Jensen, P., J.-B. Rouquier, N. Ovtracht, and C. Robardet. Characterizing the Speed and Paths of Shared Bicycle Use in Lyon. *Transportation Research Part D: Transport and Environment*, Vol. 15, No. 8, Dec. 2010, pp. 522–524.
28. Rolling Orange Bikes. Used Urban Arrow Cargo Bike, 2015. <http://rollingorangebikes.com/urban-arrow-cargo-bike/>.
29. Maus, J. New Study Compares Bicycling's CO₂ Emissions to Other Modes. *BikePortland*, Dec. 12, 2011. <http://bikeportland.org/2011/12/12/new-study-compares-bicyclings-co2-emissions-to-other-modes-63536>.
30. U.S. Government Accountability Office. *Strategy Needed to Address Aging Delivery Fleet*. United States Postal Service, 2011. <http://www.gao.gov/assets/320/318032.pdf>.
31. Minnesota Department of Transportation. *The Per-Mile Costs of Operating Automobiles and Trucks*. St. Paul, 2003. <http://conservancy.umn.edu/bitstream/handle/11299/909/1/200319.pdf>.
32. Seedah, D. P. K., J. C. Muckelston, and R. Harrison. Truck Use on Texas Toll Roads. *Journal of the Transportation Research Forum*, Vol. 52, no. 1, 2013, pp. 83–95. http://www.trforum.org/journal/downloads/2013v52n1_05_TexasTollRoads.pdf.
33. Nissan Newsroom Europe. First Zero-Emission Deliveries in Italy by Nissan e-NV200 and DHL. Feb. 5, 2015. <http://www.newsroom.nissan-europe.com/EU/en-gb/Media/Media.aspx?mediaid=129301>.
34. *Light-Duty Automotive Technology, Carbon Dioxide Emissions, and Fuel Economy Trends: 1975 Through 2014*. U.S. Environmental Protection Agency, 2014.
35. City of Austin GIS. GIS/Map Downloads. Austin Texas Government, 2015. ftp://ftp.ci.austin.tx.us/GIS-Data/Regional/coa_gis.html.
36. Cordeau, J.-F., G. Laporte, M. W. P. Savelsbergh, and D. Vigo. Chapter 6: Vehicle Routing. *Handbook in OR and MS*, Vol. 14, 2007, pp. 367–413. <http://dis.unal.edu.co/~gjhernandezp/TOS/ROUTING/VRP1.pdf>.
37. Schrimpf, G., J. Schneider, H. Stamm-Wilbrandt, and G. Dueck. Record Breaking Optimization Results Using the Ruin and Recreate Principle. *Journal of Computational Physics*, Vol. 159, 2000, pp. 139–171. <http://www.staff.uni-mainz.de/schneidj/papers/ruinandrecreate.pdf>.
38. Misevicius, A. *Ruin and Recreate Principle Based Approach for the Quadratic Assignment Problem*. Springer-Verlag, Berlin and Heidelberg, 2003, pp. 598–609.
39. GitHub. Network Based VRPs. jsprit, Aug. 8, 2013. <https://github.com/jsprit/jsprit/wiki/Network-Based-VRPs>.
40. Mehta, T. *VBA Code to Find Multiple Matching Combinations*. MS Excel and Office Consulting Business Process and Operations Strategy, Dec. 27, 2012. http://www.tushar-mehta.com/excel/templates/match_values/index.html#Solver_Template.
41. VTPI. Transportation Cost and Benefit Analysis II. 2013. <http://www.vtpi.org/tca/tca04.pdf>.

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